

Data Dictionary

Online Retail Customer Segmentation Dataset

Column Name	Data Type	Description
InvoiceNo	Text	Unique identifier for each transaction.
StockCode	Text	Unique identifier assigned to each product.
Description	Text	Name or description of the purchased product.
Quantity	Integer	Number of units purchased in a transaction.
InvoiceDate	Date/Time	Date and time the order was placed.
UnitPrice	Decimal	Price per unit of the product (£).
CustomerID	Integer	Unique identifier for each customer.
Country	Text	Customer's country of residence.
Sales	Decimal	Calculated field: $\text{Quantity} \times \text{UnitPrice}$.
Recency	Integer	Number of days since the customer's most recent purchase. Lower values indicate more recent activity.
Frequency	Integer	Total number of unique purchases made by a customer.
Monetary	Decimal	Total amount spent by the customer across all purchases.
R_Score	Integer	Score assigned based on Recency ranking (typically 1–5).
F_Score	Integer	Score assigned based on Frequency ranking (typically 1–5).
M_Score	Integer	Score assigned based on Monetary ranking (typically 1–5).
RFM_Score	Text	Combined RFM score (e.g., 555, 443) used for customer segmentation.
Customer Segment	Text	Business classification derived from RFM Score (e.g., Champions, Loyal Customers, Potential Loyalists, At Risk, Lost Customers).

Calculated Metrics

Metric	Formula
Sales	$\text{Quantity} \times \text{UnitPrice}$
Recency	$\text{Reference Date} - \text{Last Purchase Date}$
Frequency	Count of Unique Invoices per Customer
Monetary	Sum of Sales per Customer
Average Order Value	$\text{Total Revenue} \div \text{Total Orders}$

Project Tools

Tool	Purpose
Microsoft Excel	Data cleaning, preprocessing, exploratory analysis, PivotTables
SQL Server	Data import, transformation, RFM analysis, customer segmentation
Power BI	Dashboard development and interactive visualization
GitHub	Project version control and portfolio presentation